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October 24, 2026

The Editor-in-Chief
Reliability Engineering & System Safety

Dear Editor,

We submit the manuscript "Evidence, not verdicts: what 479,547 public PX4 flight logs can and cannot tell a drone fleet operator" for consideration as a research article.

Flight logs are increasingly used to decide why a small drone crashed and who was responsible. We tested that use on the largest public archive of real autopilot logs. Every confirmatory test was committed to version control, with its threshold, before the data were analysed, and decisive tests were repeated on held-out data. The results are of direct relevance to the journal's readership in safety and reliability:

- neither a declared rule nor a cross-validated model could separate pilot-caused from aircraft-caused crashes;
- checks that appear to diagnose a fault can owe their accuracy to the crash itself (a form of data leakage);
- the flight that crashed was no different from the same drone's earlier flights, whereas drones that later crashed vibrated more than matched drones on ordinary flights, a result that survived correction for multiple testing and several robustness checks;
- small language-model review panels performed worse than a fixed rule and could be steered by one compromised reviewer, and matching a written problem description against expert-diagnosed crashes named the cause less often than the base rate;
- the findings are carried into the design of an open-source fleet-assurance tool (evidence timeline, vibration ranking, signed record, three-dimensional replay and rehearsal), whose parts were checked with injected faults and a leave-one-out test, and which presents evidence rather than verdicts.

Together these findings separate what logs can support (ranking airframes for inspection) from what they cannot (attributing an individual crash), and they give practical guidance for evaluating log-based diagnostics. Code, declarations, derived features and labels are released.

The manuscript is original, has not been published and is not under consideration elsewhere. All authors approve the submission. An AI assistant was used in preparing code and text, as declared

in the manuscript.

Yours sincerely,

Dr. Nagesh Chougule
on behalf of all authors
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